MOHAMMAD BILAL AZIZ

A hands-on ML engineer who takes ownership from problem discovery to production deployment

+49 177 2049774 mbilalaziz.01@gmail.com Github LinkedIn Medium

EDUCATION

Master's of Science, Artificial Intelligence **Currently Enrolled**

- Friedrich-Alexander-Universität Erlangen-Nürnberg

Bachelor's of Science, Computer Science **Aug 2020 - June 2024**

- National University of Computer and Emerging Terchnologies
- CGPA: 3.93/4.0 (Bronze Medal)

TECHNICAL SKILLS

ML / CV: PyTorch, TensorFlow, HuggingFace Transformers, SegFormer, Federated Learning (Flower, FedML, Substra)
Data & Annotation: Segments.ai, Dataset Curation, Preprocessing Pipelines, Quality Checks, NumPy, Pandas
LLM & Agents: OpenAI API, Claude, LangChain, LlamaIndex, Multi-Agent Systems, Prompt Engineering
Infra & Cloud: AWS EC2/Neptune, GCP, Docker, Kubernetes, ElasticSearch, DuckDB, Redis, Postgres
Dev & Backend: Python, Flask, FastAPI, SQL, Linux/Unix, EFK/Kibana

PROFESSIONAL EXPERIENCE

SECURITI.AI : SOFTWARE DEVELOPER **June 2024 - Present**

- Designed and deployed a **3-agent adversarial PR evaluation system** (Reviewer · Devil's Advocate · Mediator) using Claude Code Agent Teams — catching **70% of bugs pre-merge**, reducing PR review cycle from **1–2 days to under 4 hours**, and achieving **org-wide rollout**.
- Engineered **4 specialized multi-agent LLM workflows** (code review, feature development, JIRA automation, extreme debugging) using custom subagents, scoped configurations, and reusable skill libraries — **boosting developer productivity by 10×** across teams.
- Re-architected graph processing pipeline to handle **10M+ records** by integrating DuckDB as a persistent cache layer offloading AWS Neptune and Elasticsearch — dramatically reducing query latency and improving multi-tenant scalability via Docker/Kubernetes orchestration.

DeDoc: CO-FOUNDER AND AI LEAD **Aug 2024 - Dec 2024**

- Co-founded DeDoc, an AI startup building **privacy-preserving federated ML infrastructure** for healthcare — enabling hospitals to collaboratively train models on fully encrypted, on-device patient data without raw data exposure
- **Benchmarked 4 federated learning frameworks** (Flower/flwr, FedML, TensorFlow Federated, Substra) across accuracy, scalability, and resource efficiency — determined Flower optimal for production **(98.75% accuracy, fastest training times)**, and Substra ideal for GDPR-sensitive healthcare environments; translated findings into architectural decisions for the MVP.

Analog Mutations : Blockchain and ML Intern **June 2023 - October 2023**

- Engineered and optimized **distributed ML** training infrastructure on **AWS EC2**, accelerating large-scale experimentation workflows and enabling **high-throughput** model training with **real-time visualization** and **monitoring pipelines**.
- Trained, fine-tuned, containerized, and deployed **10+ generative AI models** — including **text-to-3D**, **text-to-speech**, **DeepFace**, and **logo generation** — using **Docker** and **GCP**, leveraging advanced hyperparameter optimization and **prompt engineering** to significantly enhance model quality and inference performance.
- Built **DecentrAI**, a decentralized **blockchain-powered AI** ecosystem integrating scalable **ML inference**, **Dockerized** deployments, and smart contract orchestration on a private **Ethereum** network, securing **1st Place** at the **Soulfest Hackathon** among competing innovation teams.

RESEARCH & PROJECTS

zkVML: Zero Knowledge Verifiable Machine Learning **August 2024**

- Led design and development as corresponding author of zkVML — a Python framework fusing Zero-Knowledge Proofs with ML, enabling cryptographically verifiable model inference for KNN, Decision Trees, and CNNs using ZoKrates.
- Peer-reviewed and presented at APAN'58 International Conference, Islamabad.

SegFormer Stamp Segmentation — Computer Vision Pipeline

- Built a **complete ML pipeline**: curated and annotated 60 images via **Segments.ai**, fine-tuned SegFormer-B0 (nvidia/mit-b0) using PyTorch with Adam optimiser over **20 epochs**, achieving **validation loss of 0.037** — published model and dataset on HuggingFace (17+ downloads, integrated into a community Space).
- **HuggingFace**: bilal01/segformer-b0-finetuned-segments-stamp-verification